



OPEN ACCESS

Sprint running mechanics are associated with hamstring strain injury: a 6-month prospective cohort study of 126 elite male footballers

Christopher Bramah ,¹ Samantha Rhodes,¹ Alexandra Clarke-Cornwell ,¹ Thomas Dos'Santos²

► Additional supplemental material is published online only. To view, please visit the journal online (<https://doi.org/10.1136/bjsports-2024-108600>).

¹School of Health & Society, University of Salford, Salford, UK
²Department of Sport & Exercise Sciences, Manchester Metropolitan University, Manchester, UK

Correspondence to

Dr Christopher Bramah;
c.a.bramah@salford.ac.uk

Accepted 11 March 2025

ABSTRACT

Objective To investigate the association between sprint running biomechanics and sprint-related hamstring strain injury (HSI) in elite male football players.

Methods This prospective cohort study recruited 126 professional male football players from eight clubs in the English football league, who were followed across a 6-month period. Maximal velocity sprint running videos (240 fps) were collected from five teams during pre-season (June to August) and three teams during the in-season period (October to March) and subsequently assessed using the Sprint Mechanics Assessment Score (S-MAS) by a single, blinded assessor. Sprint-related HSI within the previous 12 months and any new MRI-confirmed sprint-related HSI were reported by club medical staff. Incidence rate ratios were calculated using a Poisson regression model to determine the association between S-MAS and new sprint-related HSIs.

Results There were 23 players with a previous sprint-related HSI and 17 new HSIs during the follow-up period, with 14 sprint-related injuries. S-MAS values were significantly greater among players with a previous HSI (median difference (MD): 1, $p=0.007$, 95% CI: 0 to 3) and those sustaining a new sprint-related HSI (MD: 2, $p=0.006$, 95% CI: 1 to 3) compared with uninjured players. Adjusting for age and previous injury found a significant association between the S-MAS and prospective sprint-related HSIs, with an adjusted incidence rate ratio of 1.33 (95% CI: 1.01 to 1.76) for each one-point increase in S-MAS.

Conclusions This is the first study to identify an association between sprint running kinematics and prospective sprint-related HSI in elite male football players. Sprint running mechanics assessed using the S-MAS were associated with both past and future HSIs, with a 33% increase in the risk of a new HSI with every one-point increase in S-MAS. Given the association to injury, evaluating sprint mechanics within rehabilitation and injury prevention may be warranted.

INTRODUCTION

Over the past two decades, hamstring strain injury (HSI) rates in professional football have continued to rise: accounting for 12% of all injuries in the 2001/2002 English football season, increasing to 24% in the 2021/2022 season.¹ This corresponds with a 10 percentage point increase in the proportion of all injury absence days due to HSI,¹ which can have significant financial and performance implications for professional football teams.² Consequently, there is a need to identify potential

WHAT IS ALREADY KNOWN ON THIS TOPIC

- ⇒ Sprint running mechanics have been suggested as a consideration in hamstring injury prevention practices.
- ⇒ Evidence of an association between sprint running mechanics and hamstring injury using prospective designs is limited.

WHAT THIS STUDY ADDS

- ⇒ This is the first study to identify sprint running mechanics to be associated with future hamstring strain injury in elite men's English Football.
- ⇒ Higher scores on the Sprint Mechanics Assessment Score (S-MAS) were associated with new hamstring injuries.
- ⇒ When adjusting for the confounding effects of previous injury and age, there was a 33% increase in hamstring strain injury rate with every one-point increase in S-MAS across a 6-month time period.
- ⇒ S-MAS scores above 5.5 had a sensitivity of 78.6% and specificity of 65.4% for new hamstring injuries.

HOW THIS STUDY MIGHT AFFECT RESEARCH, PRACTICE OR POLICY

- ⇒ The time-efficient nature of the S-MAS means this method can be integrated into regular in-season screening practices to evaluate sprint kinematics as part of performance assessments, injury screening and rehabilitation.
- ⇒ Screening sprint mechanics within rehabilitation and injury prevention may be beneficial considering their association with new hamstring injuries, although the modification of prevention programmes based on the S-MAS requires further study.
- ⇒ Given the multifactorial nature of injury development, sprint mechanics should be assessed within holistic screening programmes alongside other factors such as physical qualities, individual characteristics and training loads.

factors that can be targeted within injury prevention practices.

Several risk factors for HSI have been identified within existing literature, including previous hamstring injury,³ age,³ eccentric hamstring muscle strength,⁴ muscle fascicle length^{3 5} and sudden



© Author(s) (or their employer(s)) 2025. Re-use permitted under CC BY-NC. No commercial re-use. See rights and permissions. Published by BMJ Group.

To cite: Bramah C, Rhodes S, Clarke-Cornwell A, et al. *Br J Sports Med* Epub ahead of print: [please include Day Month Year]. doi:10.1136/bjsports-2024-108600

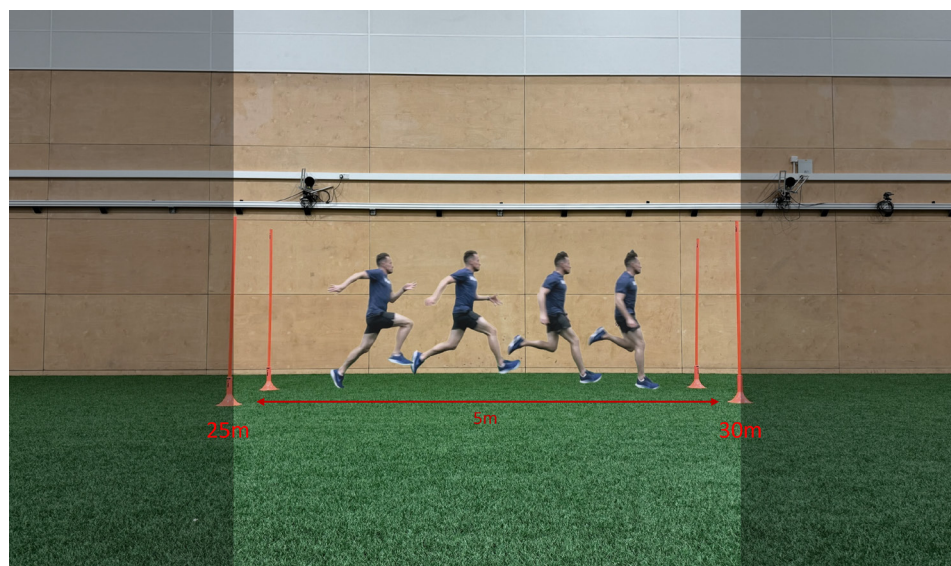


Figure 1 Diagram showing the camera setup and the perspective view from the camera of the running capture volume.

increases in sprint running exposure.⁶ As sprint running is considered the primary mechanism for HSI,¹⁷ the role of biomechanics in injury development has been widely debated.⁸ Conditioning coaches, physiotherapists^{9 10} and expert opinion articles frequently recommend modifying sprint mechanics within injury prevention and rehabilitation programmes^{8 11–14}; however, the available prospective evidence to support the association between sprint mechanics and HSI remains inconclusive.

Biomechanical factors are thought to influence HSI by altering the mechanical stress and/or strain applied to the hamstrings.¹⁵ While kinetic factors have been linked to HSI,^{16 17} the majority of existing literature has focused on sprint kinematics. Kinematic features such as increased anterior pelvic tilt,¹⁸ thoracic side flexion during swing,^{18 19} along with altered neuromuscular activity of trunk and pelvis musculature^{20 21} have been observed in individuals sustaining future HSIs. However, these studies are frequently limited by small sample sizes and a reliance on three-dimensional motion capture (3DMoCap) technology. While considered the gold standard for biomechanical assessments, 3DMoCap is difficult to implement in most team-sport settings due to the high equipment cost, time-consuming nature of assessments and the need for dedicated laboratory space. Consequently, this limits the practical ability to screen athletes as part of injury risk mitigation and rehabilitation processes.

The Sprint Mechanics Assessment Score (S-MAS) is a 12-item qualitative movement screening tool developed for the in-field assessment of kinematic features associated with HSI.²² The score uses slow-motion video analysis to evaluate the overall movement quality of an individual's sprint running mechanics; with higher scores indicating suboptimal movement patterns. Although its validity against 3DMoCap has not yet been established, the S-MAS has demonstrated good intertester and intratester reliability (intraclass correlation coefficients (ICCs) of 0.799 and 0.828, respectively).²² The practical nature of this assessment tool means that it could potentially be integrated into clinical practice to identify individuals who demonstrate biomechanical patterns associated with HSI. However, the association between the S-MAS and HSI is not yet known.

The aim of this study was to investigate the association between sprint running kinematics, using the S-MAS, and both retrospective and prospective HSI in elite male football players.

METHODS

Study design

This prospective cohort study was conducted between June 2022 and March 2024 with clubs followed over a 6-month period following initial enrolment. Methodological reporting was completed in accordance with the Strengthening the Reporting of Observational Studies in Epidemiology checklist.²³

Participants

Outfield players at professional football clubs across various leagues within the English Football League system were eligible for participation within this study. To be included, participants were required to be medically cleared for full training and match play at the start of the study and provide written informed consent. Participants were excluded if they were under the age of 18 or returning from surgery within the previous 6 months. Goalkeepers were excluded due to their limited exposure to sprint running. Football clubs within the professional network of the authors were invited to participate in this study. Nine clubs agreed, with one excluded as players did not meet the age criteria.

Data from one club collected in the first round of data collection were used within a power calculation to estimate sample size required for a dichotomous outcome²⁴ (G*Power, V.3.1): 100 participants were needed to achieve a 90% power at type 1 error (two-sided α) of 5%, based on an injury prevalence of 22% in the pilot data and an enrolment ratio of 1:4 (injured: uninjured).

Sprint running data collection

Sprint running data were collected during two 35 m maximal effort sprints from all participants. Data from five teams were collected during the preseason period (June to August) and three teams in-season (October to March). Participants first completed a 15–20 min warm-up led by the club's sports science team, followed by submaximal sprint runs at 80% and 90% of maximum effort. Two maximum effort 35 m sprint running trials were then recorded using a slow-motion camera sampling at 240 fps (iPhone 13 pro, Apple). All running trials were completed on synthetic artificial field-turf or grass football pitches, with

participants wearing their own sport-specific footwear. Witty Photocell timing gates (Microgate, USA) were positioned between 25 and 30m to monitor maximal running speed. The camera was positioned on a tripod 0.8m high and 7m perpendicular to the capture volume (figure 1). Two sprint trials were collected to ensure a clear video was obtained of the right and left limbs, respectively.

S-MAS scoring

Sprint running trials were assessed against the S-MAS by a single blinded assessor (biomechanist with 10 years experience). The S-MAS is a 12-item composite score, evaluating movement patterns across the gait cycle against specific criteria²² (online supplemental file 1). Videos were viewed using computer software Kinovea (V0.9.5) allowing footage to be played in slow motion and paused at appropriate points of the gait cycle. Using a dichotomous scoring system, players are scored one point for the presence and zero points for the absence of each of the 12 kinematic features. Scores are summed to produce a total rating of overall movement quality, with scores of zero indicating optimal mechanics and 12 for suboptimal. The S-MAS has been shown to have good intertester (ICC: 0.799, 95% CI: 0.642 to 0.892) and intratester reliability (ICC: 0.828, 95% CI: 0.688 to 0.908), with a SE of measurement of one point.²²

Hamstring injury reporting

HSIs were included in both the retrospective and prospective data analysis providing they were non-contact and sustained during sprinting in either training or match-play.

For the retrospective analysis, players self-reported any hamstring injury within the previous 12 months, the affected limb (left, right, bilateral), and injury mechanism (sprint running, tackle, kicking, stretch, other), which were subsequently confirmed by club medical staff.

For the prospective analysis, clubs were followed over a 6-month period, reporting monthly occurrence of any new HSIs. Prospective injuries were diagnosed by club medical staff and radiologically confirmed via MRI. Club staff reported details including: injury mechanism, situation (training or game-play), affected limb, muscle injured (biceps femoris, semitendinosus, semimembranosus), and the grade of each HSI classified using the British Athletics Muscle Injury Classification System.²⁵ At the end of the 6-month follow-up period, clubs were contacted to discuss if any of the players sustained any severe time-loss injury that was not a HSI, defined as an injury resulting in absence from training and games for more than 28 days, or repeated bouts (two or more) of moderate injuries ranging between eight and 28 days, based on published injury severity ratings.²⁶ These players were subsequently excluded from the uninjured group in the final prospective analysis. This exclusion criterion aimed to remove participants with reduced training exposure due to the potential confounding effect this may have on both sustaining and not sustaining a future HSI.

Equality, diversity and inclusion

Our author group is composed of individuals of diverse genders, ethnic groups and professional backgrounds including both male and female authors, senior and early career researchers. Professional backgrounds include a physiotherapist, sports scientist, statistician and a strength and conditioning coach. While the study focused on a male population, efforts were made to recruit female clubs of a similar level. However, challenges such as refusal

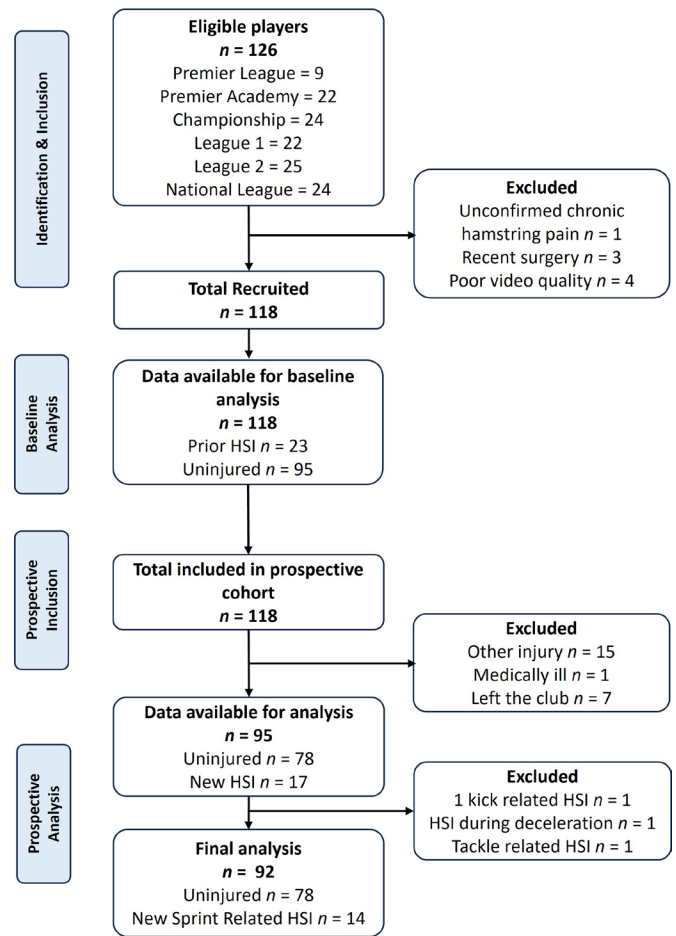


Figure 2 Flow diagram depicting participant flow through the study. HSI, hamstring strain injury.

to participate, lack of response and limited access to medical and MRI facilities for HSI verification limited participation.

Patient and public involvement

Patient and public involvement occurred throughout the design and delivery of this research. Specifically, practitioners working in football were involved in establishing the research priority and question with feedback and consultation sought to develop and refine the outcome measures used and research methodology.

Statistical analysis

Statistical analysis was performed in Stata (release 13, StataCorp) with figures created in JASP (V0.18.1, University of Amsterdam), aligning with the CHAMP statement.²⁷ Data normality was assessed using the Shapiro-Wilk test and Q-Q plots, and homogeneity of variance assessed using Levene's test. Differences between participant characteristics for uninjured and injured groups were determined using independent t-tests (normally distributed variables) and Mann-Whitney-U tests (non-normally distributed variables) in both retrospective and prospective analysis.

For the retrospective analysis, differences in S-MAS values were compared between uninjured players (PREV-UNINJ) and players with a sprint-related HSI in the previous 12 months (PREV-INJ) using Mann-Whitney-U tests. In the prospective analysis, new MRI-confirmed sprint-related HSIs were considered the primary outcome with S-MAS the independent variable. Participants who sustained a new sprint-related HSI were included in the

Table 1 Mean (SD) for normally distributed variables; median (IQR) for non-normally distributed variables; T-test for normally distributed variables; Mann-Whitney U test for non-normally distributed variables

Variable	Total (n=118)	PREV-INJ (n=23)	PREV-UNINJ (n=95)	Between group difference (95% CI)	P value
Age (years)*	21 (19, 25)	20 (19, 25)	21 (18, 25)	1 (–1 to 2)	0.474
Mass (kg)	78.8 (7.7)	79.5 (5.9)	78.4 (8.0)	–1.1 (–4.8 to 2.6)	0.556
Height (cm)	182.2 (6.0)	180.4 (6.4)	182.6 (5.8)	2.2 (–0.5 to 5.0)	0.109
Maximal running speed (m/s)*	8.9 (8.3, 9.3)	8.8 (8.3, 9.4)	8.9 (8.3, 9.3)	0 (–0.4 to 0.4)	0.984
S-MAS*	5 (2, 7)	6 (6, 7)	5 (2, 7)	1 (0 to 3)	0.007†

95% CIs were calculated using the Hodges-Lehmann estimates.

*Denotes non-normally distributed variables, assessed using Mann-Whitney U test.

†Statistical significance at $p < 0.05$.

S-MAS, Sprint Mechanics Assessment Score.

injured group (PROSP-INJ) with Mann-Whitney-U tests used to compare S-MAS values between this group and those who remained uninjured throughout the 6-month follow-up period (PROSP-UNINJ). S-MAS scores for injured players were taken from the video positioned on the same side of the injured limb and compared with a randomly selected limb of uninjured players using a random cell selector function in Microsoft Excel.

Effect sizes were calculated with Hedges' g and interpreted as 0.2 small, 0.5 medium and 0.8 large.²⁸ The Kruskal-Wallis test with post hoc Dunn's correction was used to compare S-MAS values between subgroups of players with a first-time sprint-related HSI, previous sprint-related HSI and uninjured players.

In the prospective analysis, the continuous S-MAS variable was entered into a Poisson regression model as a predictor variable for new sprint-related HSIs (outcome variable).²⁹ Age and previous hamstring injury were included in an adjusted model as potential confounders based on a prior systematic review identifying these variables as having the strongest association to prospective HSI.³ A Poisson regression model was selected to estimate adjusted incidence rate ratios (IRRs) for sprint-related HSIs over the 6-month follow-up period, as it is well suited for

modelling the incidence of discrete events within a fixed time frame.

Receiver operator characteristic (ROC) curves were used to determine a cut-off threshold for the S-MAS, defined as the value yielding the best combined sensitivity and specificity.³⁰ This categorical variable was entered into a separate Poisson regression model to determine the association between S-MAS as a categorical variable and prospective sprint-related HSI.

RESULTS

Population

A total of 126 professional male football players agreed to participate from eight clubs in the English Football League; including three Premier League (first team $n=9$, academy $n=22$), one Championship ($n=24$), two League 1 ($n=22$), one League 2 ($n=25$) and one National League ($n=24$) club. Figure 2 illustrates participant flow throughout the study. In the retrospective analysis, 118 players were included with 23 players in the PREV-INJ group (height: 180.4 cm, mass: 79.5 kg) and 95 in the PREV-UNINJ group (height: 182.6 cm, mass: 78.4 kg). For the prospective cohort analysis, 118 participants were followed across a 6-month period, 7 were lost to follow-up (5.9%) with 16 (13.5%) players excluded from the uninjured group due to sustaining a severe time loss injury or repeated bouts (two or more) of moderate injuries or illness which were not HSIs. 17 new HSIs occurred across the follow-up period, 14 of which were sprint-related. The three non-sprint-related HSIs were excluded from the final analysis, leaving a total of 92 participants with 14 participants in the PROSP-INJ group (height: 182.1 cm, mass: 79.1 kg) and 78 in the PROSP-UNINJ group (height: 181.9 cm, mass: 78.1 kg).

Retrospective analysis

Of the 118 participants included, no significant differences were found for age, mass, height or maximal running speed between the PREV-INJ group ($n=23$) and the PREV-UNINJ group ($n=95$) (table 1). Significant, trivial to large differences were observed between groups for the S-MAS ($p=0.007$, Hedges' $g=0.17$ to 1.1), with median score of five in the PREV-UNINJ and six in the PREV-INJ group.

Prospective analysis

Table 2 provides details on the frequency, mechanism and grade of all 17 HSIs sustained during the follow-up period. 92 players were included in the final analysis, with significant small to large differences observed between PROSP-UNINJ ($n=78$) and PROSP-INJ ($n=14$) groups for the S-MAS ($p=0.006$, $g=0.22$ to 1.37) (figure 3), with a median S-MAS of four in

Table 2 Hamstring injury details for all injuries including player position, injury location, grade, injury scenario and the mechanisms of injury

Injury number	Player position	Injury location	BAMIC grade	Injury scenario	Mechanism of injury
1	CM	LHBF	3c	Training	Sprint
2	FW	LHBF	2c	Training	Sprint
3	CM	BFSH	2b	Game	Sprint
4	CM	LHBF	3c	Game	Sprint
5	FB	ST	2b	Training	Sprint
6	CD	LHBF	2c	Game	Sprint
7	FB	LHBF	2b	Training	Sprint
8	CM	LHBF	3b	Game	Sprint
9	FW	LHBF	2b	Game	Stretch/kick
10	FW	LHBF	1b	Game	Sprint
11	CM	LHBF	1b	Game	Deceleration
12	CM	LHBF	4c	Game	Sprint
13	CM	LHBF	2c	Game	Sprint
14	CM	LHBF	2b	Game	Tackle
15	CD	ST	2b	Training	Sprint
16	FW	LHBF	1b	Game	Sprint
17	CM	LHBF	2b	Training	Sprint

BAMIC, British Athletics Muscle Injury Classification System; BFSH, biceps femoris short head; CD, central defence; CM, central midfield; FB, full back; FW, forward; LHBF, long head biceps femoris; ST, semitendinosus.

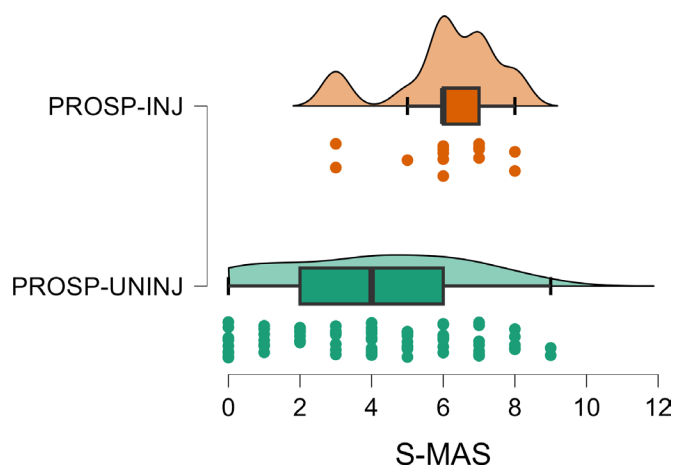


Figure 3 Raincloud plot illustrating the distribution of S-MAS values among prospectively injured participants (PROSP-INJ) and uninjured (PROSP-UNINJ). Raw data points depict individual scores, a density estimate depicts the distribution of data and boxplots depict the median (solid central line), IQR (upper and lower box) and upper and lower limit of the IQR (whiskers). Data points outside of the whiskers represent outliers. S-MAS, Sprint Mechanics Assessment Score.

the PROSP-UNINJ group compared with six in the PROSP-INJ group (table 3). Comparisons between players with a first-time sprint-related HSI (n=8) compared with PROSP-UNINJ revealed a statistically significant difference of three S-MAS points (median: 7 vs 4, $p=0.017$) (figure 4).

The continuous S-MAS variable was significantly associated with a higher rate of new sprint-related HSIs, with an IRR of 1.38 (95% CI: 1.06 to 1.79; $p=0.017$). After adjusting for age and previous HSI, the IRR remained significant at 1.33 (95% CI: 1.01 to 1.76; $p=0.044$). Indicating that each one-point increase in S-MAS corresponded to a 33% increase in the risk of developing a new sprint-related HSI during the follow-up period.

ROC curves indicated S-MAS values of 5.5 produced the highest combined sensitivity (78.6%) and specificity (65.4%) with an area under the curve of 0.732. Therefore, players were categorised into groups with a score of ≥ 6 or with scores of ≤ 5 to calculate IRRs for S-MAS as a categorical variable. The IRR for a new sprint-related HSI associated with the categorical S-MAS variable was non-significant ($p=0.065$, IRR 2.8, 95% CI: 0.94 to 8.35).

DISCUSSION

Main findings

The main findings of the present study confirm that suboptimal running kinematics, indicated by higher S-MAS values, which reflect lower sprinting movement quality, are associated

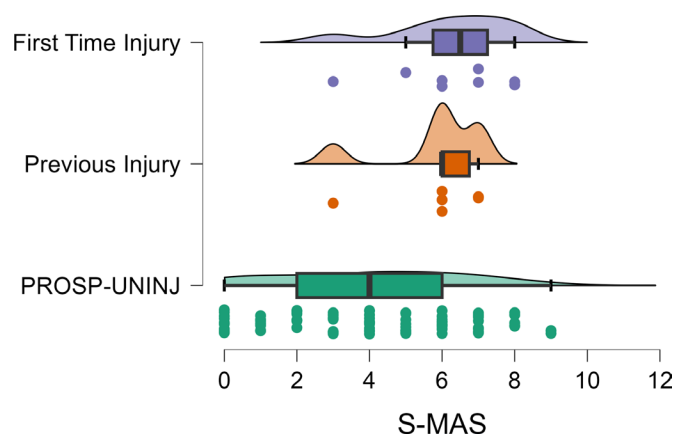


Figure 4 Raincloud plot illustrating the distribution of S-MAS values among subgroups within the prospective cohort. Previous injury includes data for individuals sustaining a new injury who had an injury in the previous 12 months. First time Injury refers to those sustaining a new hamstring injury. Raw data points depict individual scores, a density estimate depicts the distribution of data and boxplots depict the median (solid central line), interquartile range (upper and lower box) and upper and lower limit of the interquartile range (whiskers). Data points outside of the whiskers represent outliers. S-MAS, Sprint Mechanics Assessment Score.

with both past and future sprint-related HSI occurrence in elite male footballers (figure 3). When adjusting for the potential confounding effects of previous HSI and age, IRRs indicate that for every one-point increase in S-MAS there was a significant 33% increase in sprint-related HSI rate across a 6 month time period, with values of 5.5 yielding a sensitivity of 78.6% and specificity of 65.4%.

Sprint running mechanics and HSIs

Mechanistically, HSIs are attributed to the interaction between applied mechanical strain and hamstring strain capacity, with running mechanics considered one factor influencing applied mechanical strain.^{15 31} While modelling and cadaver studies support the theoretical link between mechanical features and hamstring strain,³²⁻³⁴ limited prospective studies have investigated their association with injury.³

Of the available evidence, several kinematic, kinetic and neuromuscular features have been linked to HSI development. These include lower maximal horizontal force production,^{16 17} reduced trunk and gluteal muscle activity during swing,²⁰ increased gluteus medius activity during stance,²¹ as well as increased anterior pelvic tilt¹⁸ and thoracic side flexion during swing.^{18 19} Our findings of greater S-MAS among

Table 3 Mean (SD) for normally distributed variables; median (IQR) for non-normally distributed variables

Variable	Total (n=92)	PROSP-INJ (n=14)	PROSP-UNINJ (n=78)	Between group difference (95% CI)	P value
Age (years)*	20 (18, 25)	21 (20, 25)	20 (18, 24)	1 (−1 to 3)	0.211
Mass (kg)	78.2 (7.4)	79.1 (7.1)	78.1 (7.5)	−1.0 (−5.3 to 3.3)	0.643
Height (cm)	182.0 (5.6)	182.1 (6.1)	181.9 (5.6)	−0.2 (−3.5 to 3.1)	0.893
Maximal running speed (m/s)*	8.9 (8.3, 9.3)	8.4 (7.1, 9.1)	8.9 (8.6, 9.4)	−0.5 (−1.5 to 0.1)	0.120
S-MAS*	5 (3, 6)	6 (6, 7)	4 (2, 6)	2 (1 to 3)	0.006†

T-test for normally distributed variables; Mann-Whitney U test for non-normally distributed variables. 95% CIs were calculated using the Hodges-Lehmann estimates.
 *Denotes non-normally distributed variables, assessed using Mann-Whitney U test.
 †Statistical significance at $p<0.05$.
 S-MAS, Sprint Mechanics Assessment Score.



Figure 5 Illustrative example of S-MAS scoring for an injured player (A) and uninjured player (B). Using S-MAS criteria, player A would score a total of 7 points out of 12, with 1 point awarded for each of the following criteria: trailing leg extension, lumbo-pelvic rotation, thigh separation at late swing, thigh separation at touch-down, shin angle and foot v centre of mass position at touch-down and vertical collapse at midstance. Player B would score 0 out of 12 points. MVP, maximal vertical projection; S-MAS, Sprint Mechanics Assessment Score.

prospectively injured individuals add to the existing evidence, further supporting the association between sprint running kinematics and HSI.

Association between S-MAS and prospective HSIs

A novel aspect of the current study is the use of the S-MAS, which is a composite score evaluating the overall quality of sprint running mechanics via 2D analysis.²² This methodology contrasts prior studies that have predominantly focused on associations between singular biomechanical parameters and HSI using 3DMoCap.^{18,19} The composite nature of the score aims to reflect the collective contribution of multiple mechanical features influencing the stress and strain applied to the hamstrings, which may not otherwise be accounted for when focusing on isolated biomechanical parameters.¹⁵ This is similar in approach to methods such as the Cutting Movement Assessment Score³⁵ and Landing Error Scoring System used for screening in anterior cruciate ligament injuries^{36,37} and aims to shift practitioner focus away from singular variables which in isolation may be insufficient to explain injury development.

When adjusting for age and previous HSI, we found a 33% increase in the risk of sprint-related HSI for every one-point increase in S-MAS. This finding suggests that sprint running mechanics, as assessed using the S-MAS, represent a factor associated with future sprint-related HSI and supports the use of the composite nature of the score; with values of 5.5 or above considered the optimal cut-off for determining those with 'suboptimal' mechanics. However, it is important to interpret this finding within the context of wider factors associated with HSI, particularly when considering the relatively low sensitivity (78.6%) and specificity (65.4%) of the S-MAS as an independent variable. While sprint running mechanics can influence the applied mechanical strain, physical and morphological factors including eccentric hamstring strength^{4,38} and muscle fascicle length⁵ are also associated with the development of future HSIs. These factors influence the capacity of the muscle to tolerate applied strain and might thus act as effect-measure modifiers in the relationship between sprint mechanics and HSI. Therefore, in both a research and clinical context, it is important that sprint mechanics are not considered in isolation, but instead within a multifactorial context for HSI.

Rehabilitation considerations

The current findings further suggest that sprint running mechanics are perhaps inadequately addressed during rehabilitation, as players with a previous sprint-related HSI were found to have greater S-MAS compared with uninjured players in the retrospective analysis. Considering higher S-MAS values were also associated with prospective injury development, it highlights a potential need to consider sprint mechanics within rehabilitation and injury prevention processes to mitigate potential risk of injury and/or reinjury. This is perhaps more pertinent when acknowledging that previous injury can lead to ongoing physical and/or structural deficits, reducing the ability of the hamstrings to withstand strain induced from suboptimal running mechanics.

Limitations

Importantly, we were unable to monitor individual training and game load demands that may have influenced exposure or lack of exposure to potential injury-inciting events. To mitigate this risk, we excluded players from the uninjured group if they had sustained repeated moderate or severe time loss injuries that were not HSIs (eg, ACL injuries). The rationale for this was that these players would have significantly reduced participation in sprint running, limiting their exposure to the mechanisms associated with HSI. As such, it was felt including them within the uninjured group could confound the results, as their reduced exposure to sprint running would lower their likelihood of sustaining a HSI. However, this exclusion may have inadvertently removed players who, despite having optimal sprint technique, remain susceptible to injury. This is particularly relevant for faster athletes, where the mechanical demands imposed on the hamstrings place them at the upper limits of their physical capacity,³⁹ increasing the risk of injury despite optimised running technique.

A further limitation of this study is that the power analysis was conducted using data that were also included in the final analysis. This approach was used to maximise participant numbers and ensure sufficient injury incidence rates. However, it is important to acknowledge that this introduces a post hoc element to the power analysis, which may inflate the precision of effect size estimates. That said, the effects of using internal data for sample size calculations in binary data are generally minimal.⁴⁰ To further address this, we have provided 95% CIs to aid interpretation

of the variability in the observed effect sizes and offer a more accurate representation of the results.

Despite demonstrating good intrarater and inter-rater reliability,²² it is important to note that the S-MAS has not yet been validated against 3DMoCap. Furthermore, in the current study, the rater was an experienced biomechanist and as such it remains unknown whether these results are generalisable to scores by less experienced users.

Finally, players included in this study were from multiple different leagues and were not all monitored over the same period of the football season. Some were followed from preseason to mid-season, and others over the latter half of the season. Consequently, injury risk may have varied between leagues and throughout the season due to non-mechanical factors such as changes in training load, match exposure, fatigue and physical conditioning practices.¹³ These fluctuations may have influenced both the exposure to, and the capacity to withstand injury inciting events, which were not fully accounted for in the study design.

Therefore, future work should consider monitoring sprint running mechanics as part of multifactorial prospective designs, accounting for exposure and the time-varying confounding effect of additional variables such as eccentric hamstring strength and muscle architecture. The results from this study are also limited to male football populations, whether the S-MAS is associated with injury in female footballers or other team sports requires further exploration.

Clinical implications

The association between S-MAS scores and future HSI suggests there may be benefit in utilising the S-MAS in the screening and evaluation of sprint running mechanics. Several authors recommend assessing and addressing running mechanics as part of holistic hamstring injury prevention and rehabilitation strategies.^{8 11 13} However, there has previously been a lack of methods available for practitioners to assess running mechanics in practice, with methods restricted to 3DMoCap which is not feasible to use in team sport settings. In contrast, the S-MAS is a quick and reliable method with parameters easily visible using slow-motion video footage (figure 5). Therefore, the S-MAS can be used in applied settings to evaluate, inform, and individualise training approaches based on sprint running mechanics.

Recent research highlights that targeted interventions can successfully modify sprint running mechanics.^{41 42} Specifically, Mendiguchia *et al*⁴² demonstrated a multimodal programme including technique training was effective in modifying mechanical features and improving sprint performance. Therefore, current methods can aid practitioners in the injury risk screening process, identifying individuals who may demonstrate 'suboptimal' movement patterns for which interventions can be targeted towards. However, whether interventions targeted towards sprint running mechanics can be successfully transferred to game situations and reduce HSI rates remains unknown and represents a logical next step in research.

CONCLUSIONS

This is the first study to investigate the association between sprint running mechanics and prospective sprint-related HSI in elite male English Football. Sprint running mechanics, assessed using the S-MAS, were found to be associated with both past and future sprint-related HSI development. When accounting for age and previous hamstring injury, there was a 33% increase in injury incidence risk for every one-point increase in S-MAS.

The practical nature of the S-MAS means that methods used within this study can be integrated into clinical practice, aiding practitioners with the screening of individuals who demonstrate mechanical patterns associated with HSI. However, practitioners should be aware that there are likely interactions with additional factors influencing both applied mechanical strain and strain capacity. Therefore, sprint running mechanics should be considered as part of multifactorial, holistic injury screening and risk mitigation practices.

X Christopher Bramah @chrisbramah, Samantha Rhodes @sammyrhodes97, Alexandra Clarke-Cornwell @dralexcc and Thomas Dos'Santos @TomDosSantos91

Contributors All authors have been active in reviewing the manuscript and have given final approval for publication. CB led the conception of the research design, delivery and writing of the manuscript. TDS was involved in the initial conception of the research design, data collection, analysis and revising the manuscript. SR was involved in the participant recruitment process, data collection and revising the manuscript. AC-C was involved in the data analysis, interpretation of the data, reviewing and revising the manuscript. CB is the principal investigator and guarantor.

Funding The authors have not declared a specific grant for this research from any funding agency in the public, commercial or not-for-profit sectors.

Competing interests None declared.

Patient and public involvement Patients and/or the public were involved in the design, or conduct, or reporting, or dissemination plans of this research. Refer to the Methods section for further details.

Patient consent for publication Consent obtained directly from patient(s).

Ethics approval This study involves human participants and was approved by University of Salford ethics committee, reference: 5583. Participants gave informed consent to participate in the study before taking part.

Provenance and peer review Not commissioned; externally peer reviewed.

Data availability statement Data sharing not applicable as no datasets generated and/or analysed for this study.

Supplemental material This content has been supplied by the author(s). It has not been vetted by BMJ Publishing Group Limited (BMJ) and may not have been peer-reviewed. Any opinions or recommendations discussed are solely those of the author(s) and are not endorsed by BMJ. BMJ disclaims all liability and responsibility arising from any reliance placed on the content. Where the content includes any translated material, BMJ does not warrant the accuracy and reliability of the translations (including but not limited to local regulations, clinical guidelines, terminology, drug names and drug dosages), and is not responsible for any error and/or omissions arising from translation and adaptation or otherwise.

Open access This is an open access article distributed in accordance with the Creative Commons Attribution Non Commercial (CC BY-NC 4.0) license, which permits others to distribute, remix, adapt, build upon this work non-commercially, and license their derivative works on different terms, provided the original work is properly cited, appropriate credit is given, any changes made indicated, and the use is non-commercial. See: <http://creativecommons.org/licenses/by-nc/4.0/>.

ORCID iDs

Christopher Bramah <http://orcid.org/0000-0003-3644-9873>

Alexandra Clarke-Cornwell <http://orcid.org/0000-0001-9510-7676>

REFERENCES

- Ekstrand J, Bengtsson H, Waldén M, *et al*. Hamstring injury rates have increased during recent seasons and now constitute 24% of all injuries in men's professional football: the UEFA Elite Club Injury Study from 2001/02 to 2021/22. *Br J Sports Med* 2022;57:292–8.
- Eliakim E, Morgulev E, Lidor R, *et al*. Estimation of injury costs: financial damage of English Premier League teams' underachievement due to injuries. *BMJ Open Sport Exerc Med* 2020;6:e000675.
- Green B, Bourne MN, van Dyk N, *et al*. Recalibrating the risk of hamstring strain injury (HSI): A 2020 systematic review and meta-analysis of risk factors for index and recurrent hamstring strain injury in sport. *Br J Sports Med* 2020;54:1081–8.
- Opar DA, Williams MD, Timmins RG, *et al*. Eccentric hamstring strength and hamstring injury risk in Australian footballers. *Med Sci Sports Exerc* 2015;47:857–65.
- Timmins RG, Bourne MN, Shield AJ, *et al*. Short biceps femoris fascicles and eccentric knee flexor weakness increase the risk of hamstring injury in elite football (soccer): a prospective cohort study. *Br J Sports Med* 2016;50:1524–35.
- Duhig S, Shield AJ, Opar D, *et al*. Effect of high-speed running on hamstring strain injury risk. *Br J Sports Med* 2016;50:1536–40.

- 7 Gronwald T, Klein C, Hoenig T, *et al.* Hamstring injury patterns in professional male football (soccer): a systematic video analysis of 52 cases. *Br J Sports Med* 2022;56:165–71.
- 8 Edouard P, Mendiguchia J, Gueix K, *et al.* Sprinting: a key piece of the hamstring injury risk management puzzle. *Br J Sports Med* 2023;57:4–6.
- 9 Freeman BW, Talpey SW, James LP, *et al.* Sprinting and hamstring strain injury: Beliefs and practices of professional physical performance coaches in Australian football. *Phys Ther Sport* 2021;48:12–9.
- 10 Kalema RN, Duhig SJ, Williams MD, *et al.* Sprinting technique and hamstring strain injuries: A concept mapping study. *J Sci Med Sport* 2022;25:209–15.
- 11 Macdonald B, McAleer S, Kelly S, *et al.* Hamstring rehabilitation in elite track and field athletes: applying the British Athletics Muscle Injury Classification in clinical practice. *Br J Sports Med* 2019;53:1464–73.
- 12 Paton BM, Read P, van Dyk N, *et al.* London International Consensus and Delphi study on hamstring injuries part 3: rehabilitation, running and return to sport. *Br J Sports Med* 2023;57:278–91.
- 13 Buckthorpe M, Wright S, Bruce-Low S, *et al.* Recommendations for hamstring injury prevention in elite football: translating research into practice. *Br J Sports Med* 2019;53:449–56.
- 14 McCall A, Pruna R, Van der Horst N, *et al.* Exercise-Based Strategies to Prevent Muscle Injury in Male Elite Footballers: An Expert-Led Delphi Survey of 21 Practitioners Belonging to 18 Teams from the Big-5 European Leagues. *Sports Med* 2020;50:1667–81.
- 15 Bramah C, Mendiguchia J, Dos'Santos T, *et al.* Exploring the Role of Sprint Biomechanics in Hamstring Strain Injuries: A Current Opinion on Existing Concepts and Evidence. *Sports Med* 2024;54:783–93.
- 16 Edouard P, Lahti J, Nagahara R, *et al.* Low Horizontal Force Production Capacity during Sprinting as a Potential Risk Factor of Hamstring Injury in Football. *Int J Environ Res Public Health* 2021;18:7827.
- 17 Edouard P, Lahti J, Fleres L, *et al.* A musculoskeletal multifactorial individualised programme for hamstring muscle injury risk reduction in professional football: results of a prospective cohort study. *BMJ Open Sport Exerc Med* 2024;10:e001866.
- 18 Schuermans J, Van Tiggelen D, Palmans T, *et al.* Deviating running kinematics and hamstring injury susceptibility in male soccer players: Cause or consequence? *Gait Posture* 2017;57:270–7.
- 19 Kenneally-Dabrowski C, Brown NAT, Warmenhoven J, *et al.* Late swing running mechanics influence hamstring injury susceptibility in elite rugby athletes: A prospective exploratory analysis. *J Biomech* 2019;92:112–9.
- 20 Schuermans J, Danneels L, Van Tiggelen D, *et al.* Proximal Neuromuscular Control Protects Against Hamstring Injuries in Male Soccer Players: A Prospective Study With Electromyography Time-Series Analysis During Maximal Sprinting. *Am J Sports Med* 2017;45:1315–25.
- 21 Franettovich Smith MM, Bonacci J, Mendis MD, *et al.* Gluteus medius activation during running is a risk factor for season hamstring injuries in elite footballers. *J Sci Med Sport* 2017;20:159–63.
- 22 Bramah C, Tawiah-Dodoo J, Rhodes S, *et al.* The Sprint Mechanics Assessment Score: A Qualitative Screening Tool for the In-field Assessment of Sprint Running Mechanics. *Am J Sports Med* 2024;52:1608–16.
- 23 Cuschieri S. The STROBE guidelines. *Saudi J Anaesth* 2019;13:S31–4.
- 24 Campbell MJ, Julious SA, Altman DG. Estimating sample sizes for binary, ordered categorical, and continuous outcomes in two group comparisons. *BMJ* 1995;311:1145–8.
- 25 Pollock N, James SLJ, Lee JC, *et al.* British athletics muscle injury classification: a new grading system. *Br J Sports Med* 2014;48:1347–51.
- 26 Ekstrand J, Häggglund M, Waldén M. Epidemiology of muscle injuries in professional football (soccer). *Am J Sports Med* 2011;39:1226–32.
- 27 Mansournia MA, Collins GS, Nielsen RO, *et al.* A Checklist for statistical Assessment of Medical Papers (the CHAMP statement): explanation and elaboration. *Br J Sports Med* 2021;55:1009.
- 28 Cohen J. Statistical power analysis for the behavioral sciences. Routledge, 2013.
- 29 McNutt LA, Wu C, Xue X, *et al.* Estimating the relative risk in cohort studies and clinical trials of common outcomes. *Am J Epidemiol* 2003;157:940–3.
- 30 Florkowski CM. Sensitivity, specificity, receiver-operating characteristic (ROC) curves and likelihood ratios: communicating the performance of diagnostic tests. *Clin Biochem Rev* 2008;29 Suppl 1:S83–7.
- 31 Duhig SJ, Bourne MN, Buhmann RL, *et al.* Effect of concentric and eccentric hamstring training on sprint recovery, strength and muscle architecture in inexperienced athletes. *J Sci Med Sport* 2019;22:769–74.
- 32 Higashihara A, Nagano Y, Takahashi K, *et al.* Effects of forward trunk lean on hamstring muscle kinematics during sprinting. *J Sports Sci* 2015;33:1366–75.
- 33 Chumanov ES, Heiderscheit BC, Thelen DG. The effect of speed and influence of individual muscles on hamstring mechanics during the swing phase of sprinting. *J Biomech* 2007;40:3555–62.
- 34 Mendiguchia J, Garrues MA, Schilders E, *et al.* Anterior pelvic tilt increases hamstring strain and is a key factor to target for injury prevention and rehabilitation. *Knee Surg Sports Traumatol Arthrosc* 2024;32:573–82.
- 35 Dos'Santos T, McBurnie A, Donelon T, *et al.* A qualitative screening tool to identify athletes with "high-risk" movement mechanics during cutting: The cutting movement assessment score (CMAS). *Phys Ther Sport* 2019;38:152–61.
- 36 Padua DA, DiStefano LJ, Beutler AI, *et al.* The Landing Error Scoring System as a Screening Tool for an Anterior Cruciate Ligament Injury-Prevention Program in Elite-Youth Soccer Athletes. *J Athl Train* 2015;50:589–95.
- 37 Smith HC, Johnson RJ, Shultz SJ, *et al.* A prospective evaluation of the Landing Error Scoring System (LESS) as a screening tool for anterior cruciate ligament injury risk. *Am J Sports Med* 2012;40:521–6.
- 38 Bourne MN, Opar DA, Williams MD, *et al.* Eccentric Knee Flexor Strength and Risk of Hamstring Injuries in Rugby Union: A Prospective Study. *Am J Sports Med* 2015;43:2663–70.
- 39 Dorn TW, Schache AG, Pandy MG. Muscular strategy shift in human running: dependence of running speed on hip and ankle muscle performance. *J Exp Biol* 2012;215:1944–56.
- 40 Coffey CS, Muller KE. Controlling Test Size While Gaining the Benefits of an Internal Pilot Design. *Biometrics* 2001;57:625–31.
- 41 Iwasaki R, Takahashi N, Shinkai H. Acute effects of sprint training for hamstrings injury prevention on male college soccer players. *sci j sport perform* 2023;3:1–9.
- 42 Mendiguchia J, Castaño-Zambudio A, Jiménez-Reyes P, *et al.* Can We Modify Maximal Speed Running Posture? Implications for Performance and Hamstring Injury Management. *Int J Sports Physiol Perform* 2022;17:374–83.